Everything in Moderation: The Currency-Maturity Composition of Sovereign Debt

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Motivation

- **How** should emerging market (EMEs) borrow?
- Two crucial dimensions of sovereign debt portfolios:
  1. **Currency**: Domestic currency (DC) vs. foreign currency (FC)
  2. **Maturity**: Long-term (LT) vs. short-term (ST)
- Conventional wisdom:
  - **Currency:**
    - DC debt provides **insurance** benefits
    - FC debt provides **disciplining** benefits
  - **Maturity**:
    - LT debt provides **insurance** benefits
    - ST debt provides **disciplining** benefits
- If the government values
  - **insurance** $\Rightarrow$ long-term & domestic
  - **discipline** $\Rightarrow$ short-term & foreign

Is DC debt of shorter maturity than FC debt in the data?

This Paper - Empirical

What I Do:

- Analyze panel data on EMEs' debt issuance and spreads

Results:

- Spreads increasing is associated with sovereigns issuing more:
  - foreign currency debt
  - short-term debt (both domestic and foreign)
- EMEs' DC debt is consistently of much lower maturity than their FC debt
  - DC average maturity: 4.7 years
  - FC average maturity: 12.5 years

This Paper - Quantitative

What I Do:

- Present a sovereign debt model with portfolio choice and calibrate to Türkiye
- Benevolent government faces uncertainty over future: Why Türkiye - currency
  1. Output
  2. Exchange rate
- Benevolent government **cannot commit** to:
  1. Repaying debt
  2. Monetary policy
  3. Borrowing decisions

Results:

- Model endogenously generates maturity gap between DC and FC debt
- Currency and maturity's insurance and disciplining properties **compound**
- Issuing long-domestic debt is too expensive in most states
- Issuing short-foreign debt is too risky in most states

Literature Review

1. **Single-bond:** Arellano (2008), Hatchondo and Martinez (2009), Chatterjee and Eyigungor (2012), Aguiar et al. (2013), Hurtado et al. (2023)
2. **Maturity Choice**: Arellano and Ramanarayanan (2012), Hatchondo et al. (2016), Bocola and Dovis (2019), Mihalache (2020)
3. **Currency Choice**: Ottonello and Perez (2019), Du et al. (2020), Sunder-Plassmann (2020), Lee (2021), Engel and Park (2022), Maeng (2025)

First to examine both aspects of sovereign debt **jointly**.

Roadmap

Empirical Analysis

Model

Quantitative Analysis

Optimal Portfolio

Lira Crisis Experiment

Empirical Analysis

Data

1. **Issuance:** Bloomberg
   - 1990 - 2024
   - 19 EMEs
   - 24,910 bonds

Data Cleaning | Issuance Summary Stats

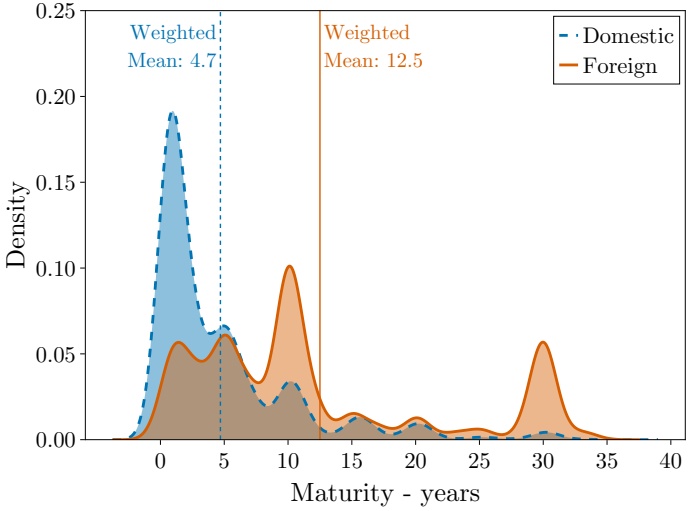
2. **Spreads:** Datastream & Du and Schreger (2016)

Spread Summary Stats

$$y^{DC}_{i,n,t} - y^{USD}_{US,n,t}$$

   - $y^{DC}_{i,n,t}$: n-year domestic currency government bond yield in country i
   - $y^{USD}_{US,n,t}$: n-year US Treasury yield
   - 12 EMEs

Distribution of Maturities



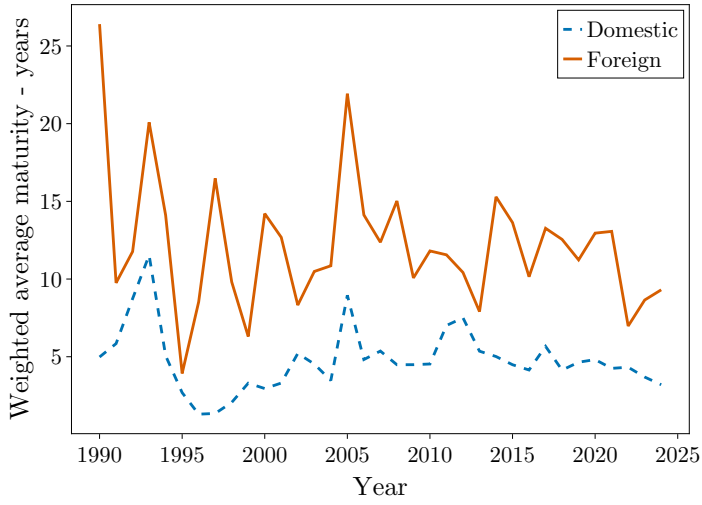
Consistent Across Countries

Country	Domestic	Foreign
Angola	3.4 years	10.6 years
Argentina	6.5 years	13.5 years
Colombia	11.2 years	15.8 years
Egypt	1.6 years	6.6 years
Hungary	4.7 years	10.6 years
Indonesia	10.0 years	14.0 years
Jamaica	8.8 years	6.1 years
Jordan	3.7 years	6.8 years
Mexico	3.8 years	18.4 years
Philippines	5.2 years	15.1 years
Poland	5.6 years	11.6 years
Qatar	4.9 years	15.3 years
Romania	4.8 years	10.2 years
Russia	6.7 years	17.5 years
Tunisia	5.7 years	9.5 years
Türkiye	3.5 years	7.5 years
Ukraine	5.7 years	6.6 years
Uruguay	11.6 years	18.0 years
Venezuela	2.6 years	17.2 years
Average	5.8 years	12.2 years
Median	5.2 years	11.6 years

- Foreign currency debt is issued at **longer** maturities than domestic currency debt
- Holds for nearly every country

Back

Consistent Across Time



Relationship Between Spreads & Portfolio Choice

▸ Binscatters ▸ Correlations

	(1) $\mathbb{1}\{\text{Domestic Currency}\}$	(2)	(3)	(4)
		Maturity (Years)		
	(Logit)	All	Domestic	Foreign
Spread	-0.023^{***}	-0.054^{**}	-0.050^{**}	-0.097^{*}
	(0.005)	(0.022)	(0.023)	(0.053)
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	12,696	1,077	1,018	561
R^2	–	0.271	0.251	0.245

Empirical Summary

- EMEs' foreign currency debt is of significantly longer maturity than their domestic currency debt
- Higher spreads is associated with EMEs tilting their portfolio towards
  1. More **foreign currency** debt
  2. More **short-term** debt (both domestic and foreign)

Model

Environment

- Small open economy model
- Three agents:
  - Risk-averse household
  - Benevolent sovereign
  - Foreign lenders
- Two sources of uncertainty, $\mathbb{X}_t = \{\mathcal{Y}_t, \mathcal{E}_t\}$:
  1. $\mathcal{Y}_t$: endowment shock, and
  2. $\mathcal{E}_t$: real exchange rate shock
- Two consumption goods - domestic and foreign

Agent 1: Households

Preferences:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left[u(C_t) - l(\Pi_t - \Pi^*) \right] \right] \tag{1}$$

- $\beta \in (0, 1)$: discount factor
- $u(\cdot)$: utility from consuming the domestic good
- $l(\cdot)$: disutility from inflation deviations
- Π^*: target inflation rate

Budget Constraint:

$$C_t = \tau_t \tag{2}$$

Agent 2: Benevolent Sovereign

- Lacks commitment in:
  1. Repaying debt
  2. Monetary policy
  3. Borrowing decisions
- 5 choice variables:
  1. Inflation
  2. 4-bond portfolio:
     - 2.1 Short-foreign: b_{t+1}^{SF}
     - 2.2 Long-foreign: b_{t+1}^{LF}
     - 2.3 Short-domestic: b_{t+1}^{SD}
     - 2.4 Long-domestic: b_{t+1}^{LD}
- $b_t > 0$: borrowing, $b_t < 0$: saving

Currency

Foreign currency:

- Sovereign receives $q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) \times b^{SF}_{t+1}$ today
- Repays b^{SF}_{t+1} tomorrow

Domestic currency:

- Sovereign receives $q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) \times b^{SD}_{t+1}$ today
- Repays b^{SD}_{t+1}/Π_{t+1} tomorrow

Maturity

Long-term debt's law of motion, following Hatchondo and Martinez (2009):

$$i_t^{LF} = b_{t+1}^{LF} - (1-\delta) b_t^{LF} \tag{3}$$

- $\delta \in (0, 1]$: fraction of LT debt that matures each period
- Sovereign receives $q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) \times (b_{t+1}^{LF} - (1-\delta) b_t^{LF})$ today
- Repays b_{t+1}^{LF} tomorrow
- **Short-term** debt: $\delta = 1$:

Sovereign's Budget Constraint

$$\overbrace{\tau_t}^{\text{Transfers}} + \overbrace{\mathcal{E}_t\left[b_t^{SF} + b_t^{LF}\right] + \frac{b_t^{SD} + b_t^{LD}}{\Pi_t}}^{\text{Debt Obligations}} =$$

$$\overbrace{\begin{aligned} & q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SD} + q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)\left[b_{t+1}^{LD} - (1-\delta)\frac{b_t^{LD}}{\Pi_t}\right] \\ & + \mathcal{E}_t\left[q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) b_{t+1}^{SF} + q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t)\left[b_{t+1}^{LF} - (1-\delta) b_t^{LF}\right]\right] \end{aligned}}^{\text{Revenue from DC Debt Issuance}} + \mathcal{Y}_t \qquad (4)$$

$$\text{Revenue from FC Debt Issuance}$$

Default

- All debt obligations erased
- No selective default
- Punishments for defaulting:
  1. Output loss:
  $$C_t = h(\mathcal{Y}_t), \quad h(\mathcal{Y}_t) \leq \mathcal{Y}_t \tag{5}$$
  2. Financial autarky: regain access with i.i.d. probability θ

Value Functions

Beginning-of-period value function:

$$V(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{D,R\}} \left\{ V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t) \right\} \tag{6}$$

Repayment policy function:

$$r(\mathbb{B}_t, \mathbb{X}_t) = \mathbb{1}\left\{ V^R(\mathbb{B}_t, \mathbb{X}_t) \geq V^D(\mathbb{X}_t) \right\}. \tag{7}$$

Value function for defaulting:

$$V^D(\mathbb{X}_t) = u(h(\mathcal{Y}_t)) + \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[\theta V(\mathbf{0}, \mathbb{X}_{t+1}) + (1-\theta) V^D(\mathbb{X}_{t+1}) \right] \tag{8}$$

Value function for repaying:

$$V^R(\mathbb{B}_t, \mathbb{X}_t) = \max_{\{\mathbb{B}_{t+1}, \Pi_t\}} \left\{ u(C_t) - l(\Pi_t - \Pi^*) + \beta \mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t} \left[V(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \right] \right\} \tag{9}$$

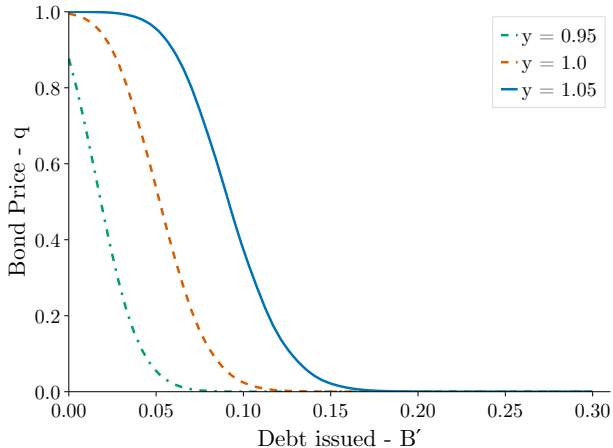
Agent 3: Foreign Lenders

- Risk-neutral,
- Perfectly competitive,
- Can access a riskless security that pays R^*

$\Longrightarrow$ break even in expectation

Bond Prices: Short-Foreign

$$q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1})\right] \tag{10}$$

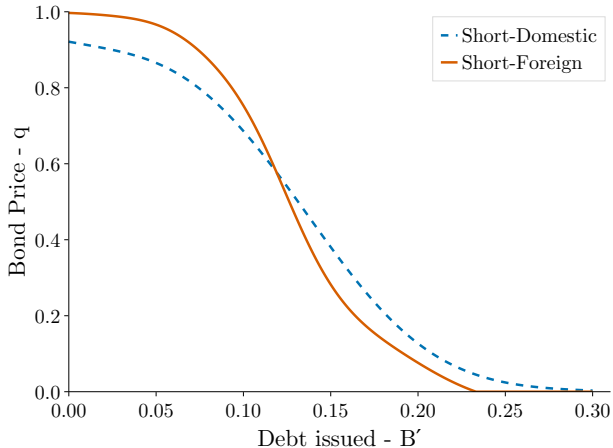


Compensation for:

1. Default risk

Bond Prices: Short-Domestic

$$q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\Pi_{t+1}}\right] \tag{11}$$

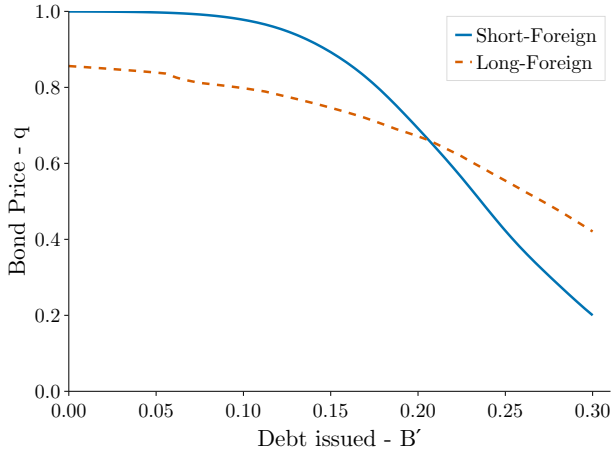


Compensation for:

1. Default risk
2. Currency risk

Bond Prices: Long-Foreign

$$q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \left[1 + (1-\delta)q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})\right]\right] \quad (12)$$

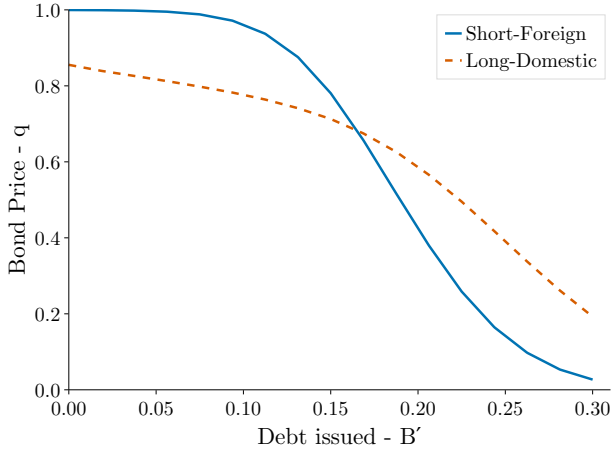


Compensation for:

1. Default risk
2. Dilution risk

Bond Prices: Long-Domestic

$$q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \times \frac{1}{\Pi_{t+1}} \times \left[1 + (1-\delta)q^{LD}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})\right]\right] \tag{13}$$



Compensation for:

1. Default risk
2. Currency risk
3. Dilution risk

Markov-Perfect Equilibrium

An MPE for this economy consists of a set of:

1. value functions $\left\{V(\mathbb{B}_t, \mathbb{X}_t), V^D(\mathbb{X}_t), V^R(\mathbb{B}_t, \mathbb{X}_t)\right\}$,
2. pricing functions $\left\{q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)\right\}$, and
3. policy functions $\left\{\hat{b}^{LD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{LF}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SD}(\mathbb{B}_t, \mathbb{X}_t), \hat{b}^{SF}(\mathbb{B}_t, \mathbb{X}_t), \hat{\Pi}_t(\mathbb{B}_t, \mathbb{X}_t)\right\}$

such that the following conditions hold:

1. Taking the bond price functions as given, the sovereign's policy functions and value functions solve the sovereign's problem, and
2. The bond price functions $q^{SF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{SD}(\mathbb{B}_{t+1}, \mathbb{X}_t), q^{LD}(\mathbb{B}_{t+1}, \mathbb{X}_t)$ satisfy (10), (12), (11), and (13), respectively.

Quantitative Analysis

Full Quantitative Model

Additional Ingredients:

1. ST bonds pay coupon: $\kappa^S = R^*$ & LT bonds pay coupon: $\kappa^L = \delta + R^* - 1$
   ▸ Pricing functions
2. Sovereign receives taste shocks to the repayment-default decision ▸ Taste shocks

$$\varepsilon^D, \varepsilon^R \sim \text{Gumbel}(-\alpha(\gamma + \ln(2)), \alpha), \quad \gamma \approx 0.5772$$

3. Household has recursive preferences Weil (1990), Epstein and Zin (1991)

Iterative Strategy:

- Backward iteration from terminal period where sovereign buys back all unmatured LT debt
- Approximate all equilibrium functions as Gaussian processes

▸ Gaussian Process Regression Details ▸ Bayesian Active Learning ▸ Iterative Strategy

Functional Forms

- **Period Utility:**

$$u(C_t) - l(\Pi_t - \Pi^*) = \frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\psi}{2}(\Pi_t - \Pi^*)^2$$

- **Output Cost of Defaulting:**

▸ Output cost

$$h(\mathcal{Y}_t) = \mathcal{Y}_t - \max\left\{0, \lambda_0 \mathcal{Y}_t + \lambda_1 \mathcal{Y}_t^2\right\}, \quad \lambda_0 < 0 < \lambda_1$$

- **Stochastic Process:**

$$\begin{pmatrix} \log(\mathcal{Y}_t) \\ \log(\mathcal{E}_t) \end{pmatrix} = \begin{pmatrix} \rho_y & 0 \\ 0 & \rho_e \end{pmatrix} \begin{pmatrix} \log(\mathcal{Y}_{t-1}) \\ \log(\mathcal{E}_{t-1}) \end{pmatrix} + \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix}, \begin{pmatrix} \varepsilon_t^y \\ \varepsilon_t^e \end{pmatrix} \sim N\left(0, \begin{pmatrix} \sigma_y^2 & \rho_{y,e}\sigma_y\sigma_e \\ \rho_{y,e}\sigma_y\sigma_e & \sigma_e^2 \end{pmatrix}\right)$$

Calibration

Maturity Currency

Parameter	Description	Value	Source
σ	risk aversion	5	Ottonello and Perez (2019)
ϱ	elasticity of intertemporal substitution	0.5	Standard Value
R^*	risk-free rate	1.04	1-year US Treasury yield
δ	perpetuity decay factor	0.1	10-year maturity
θ	re-entry probability	0.17	6-year exclusion
ρ_y	output persistence	0.728	Data
σ_y	output volatility	0.028	Data
ρ_e	real exchange rate persistence	0.879	Data
σ_e	real exchange rate volatility	0.096	Data
$\rho_{y,e}$	shock correlation	-0.326	Data
β	discount factor	0.86	Moment Matching
λ_0	default cost - linear	-0.481	Moment Matching
λ_1	default cost - quadratic	0.516	Moment Matching
ψ	inflation disutility	1.1	Moment Matching
Π^*	inflation target	1.083	Moment Matching

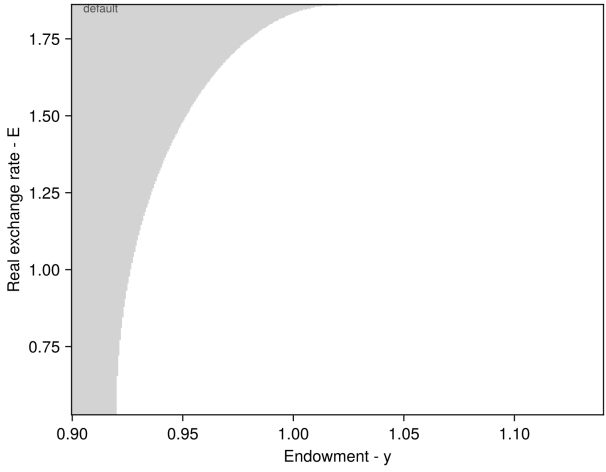
Model Fit - Targeted Moments

Statistics	**Data**	**Model**
Mean debt-to-GDP ratio	12.6%	10.9%
Mean spread (1 year, domestic currency)	10.3%	9.6%
Std. spread (1 year, domestic currency)	3.0%	2.1%
Mean inflation	8.6%	8.7%
Std. inflation	1.8%	1.4%

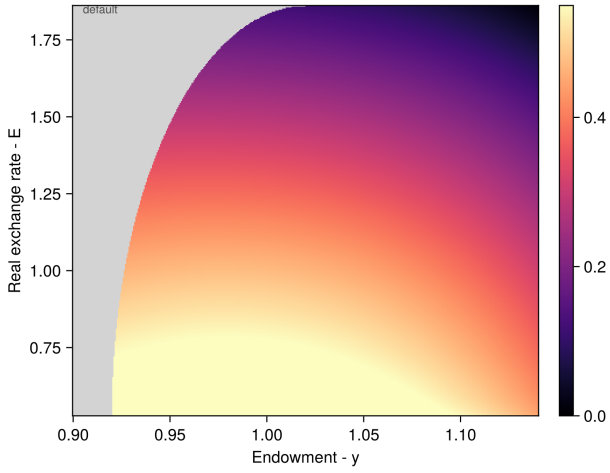
Model Fit - Untargeted Moments

Statistics	**Data**	**Model**
Domestic currency share	89.7%	81.7%
Mean maturity (domestic currency)	3.1 years	2.2 years
Mean maturity (foreign currency)	8.6 years	5.8 years
Mean spread (1 year, foreign currency)	1.49%	1.23%
Mean spread (10 year, foreign currency)	4.74%	3.3%
Mean spread (10 year, domestic currency)	7.2%	9.9%

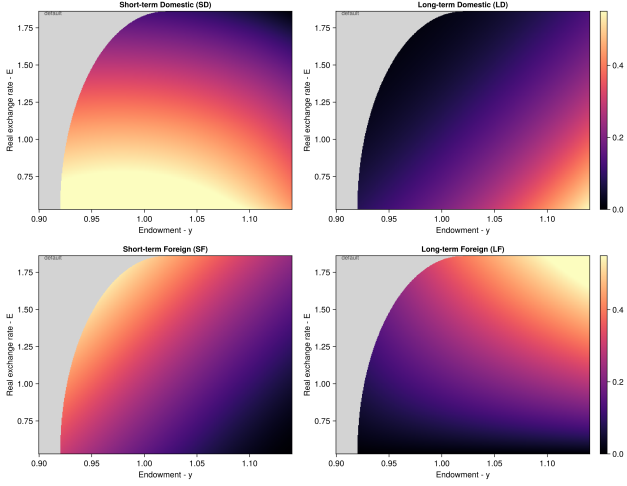
Optimal Portfolio Across States



Optimal Portfolio - Short Domestic



Optimal Portfolio Across States



Optimal Portfolio

Generalized Euler Equations

- Each bond's first-order necessary optimality condition:

$$u_C(C_t) \times [\text{Marginal Revenue}] = \beta \mathbb{E}_t \left[u_C(C_{t+1}) \times \text{Real Cost of Repayment} \right] \tag{14}$$

- Bonds differ in their ability to:
  - Raise revenue for the sovereign at period t
  - Insure the sovereign against future risk at period $t+1$

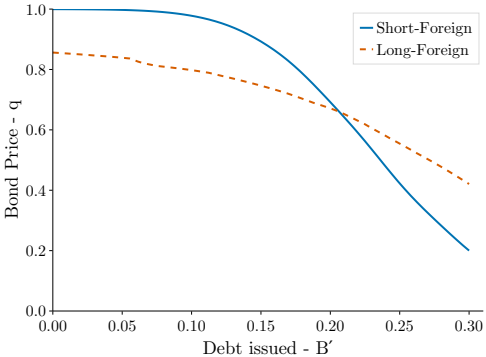
Long-Term Debt's Incentive Costs

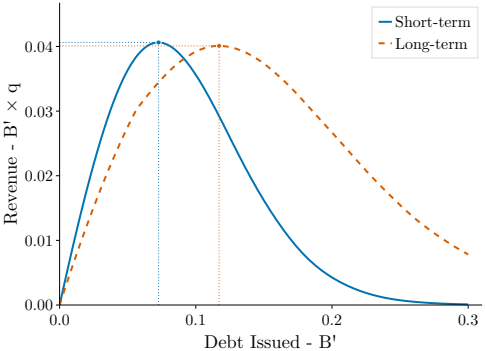
- Price of long-term debt:

$$q^{LF}(\mathbb{B}_{t+1}, \mathbb{X}_t) = \frac{1}{R^*}\mathbb{E}_{\mathbb{X}_{t+1}|\mathbb{X}_t}\left[r(\mathbb{B}_{t+1}, \mathbb{X}_{t+1}) \times \left[\kappa^L + \underbrace{(1-\delta)q^{LF}(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})}_{\text{market value of unmatured debt}}\right]\right]$$

- Sovereign lacks commitment over $\mathbb{B}_{t+2}$/future borrowing choice
- Investors price in this risk

Long-Term Debt's Incentive Costs





Long-Term Debt's Hedging Benefit

- Real cost of repaying 1 unit of short-term debt:

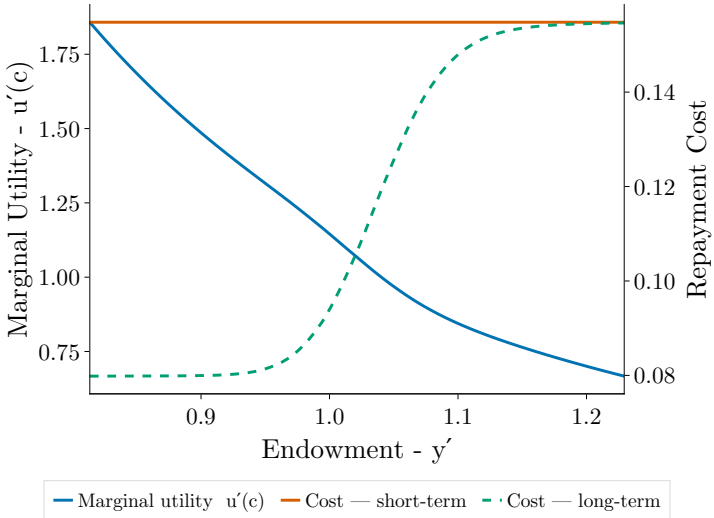
$$\kappa^S$$

- Real cost of repaying 1 unit of long-term debt:

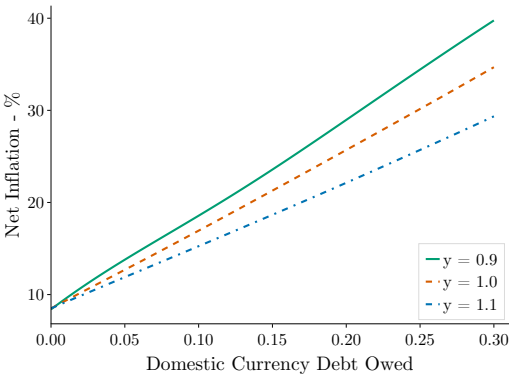
$$\kappa^L + (1-\delta)q^L(\mathbb{B}_{t+2}, \mathbb{X}_{t+1})$$

- Cost of repaying long-term debt varies with the state of the world

Long-term Debt's Hedging Benefit



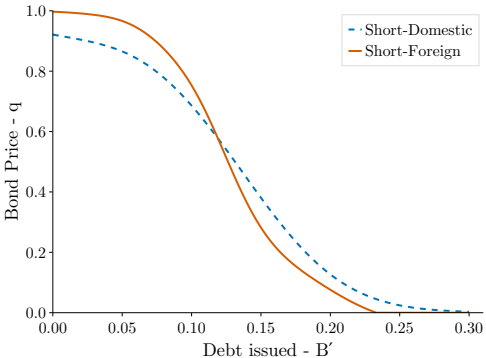
Domestic Currency Debt's Incentive Cost

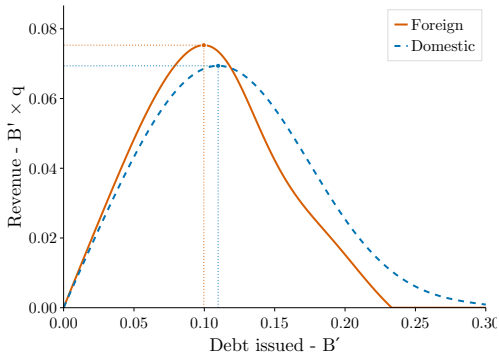


- Incentive to inflate tomorrow increases with domestic currency debt owed
- Investors price in this risk

Domestic Currency Debt's Incentive Cost

▸ Comparative statics





Domestic Currency Debt's Hedging Benefit

- Real cost of repaying 1 unit of foreign currency debt:

$$\mathcal{E}_{t+1}$$

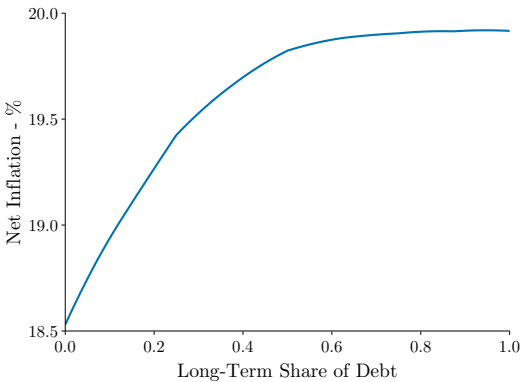
- $\text{corr}(\mathcal{E}_{t+1}, \mathcal{Y}_{t+1}) < 0$
- $\text{corr}(1/\hat{\Pi}_{t+1}, \mathcal{Y}_{t+1}) > 0$
- Real cost of repaying 1 unit of domestic currency debt:

$$\frac{1}{\hat{\Pi}_{t+1}(\mathbb{B}_{t+1}, \mathbb{X}_{t+1})}$$

Long-Term Domestic Currency Debt's Incentive Cost

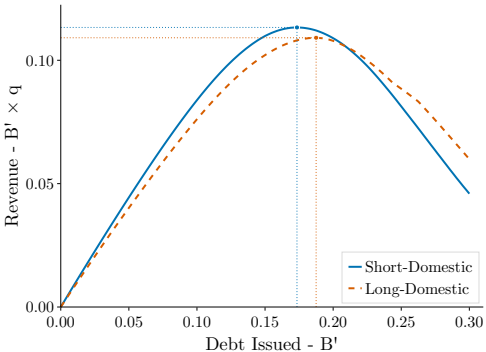
Comparative statics

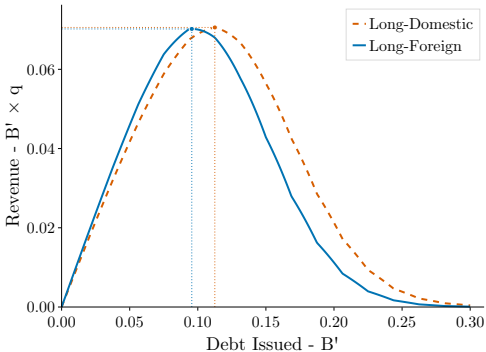
$$u_C(C_t)\left[\frac{1}{\Pi_t^2}\right]\left[\kappa^L b_t^{LD} + \kappa^S b_t^{SD} + \underbrace{q_t^{LD}(1-\delta)b_t^{LD}}_{\text{Unmatured LD Debt}}\right] = l_\Pi(\Pi_t - \Pi^*) \qquad (15)$$



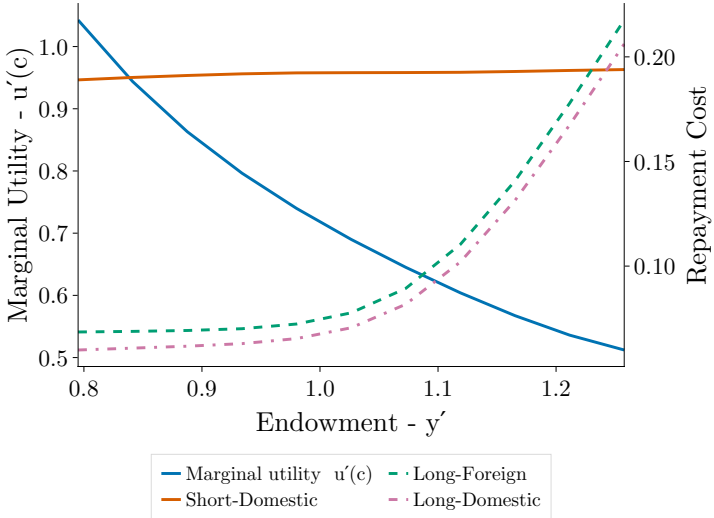
- Long-term debt increases inflation incentives **more** than short-term
- The longer-term the LD bond, the greater the incentive to inflate

Long-Term Domestic Currency Debt's Incentive Costs

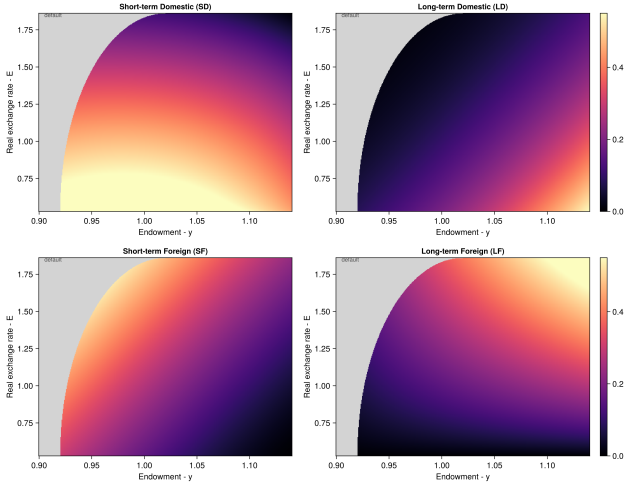




Domestic Currency Debt's Hedging Benefit

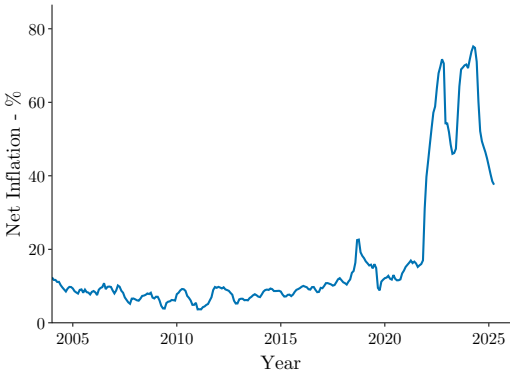


There Is A Time For Everything



Lira Crisis Experiment

Experiment



- In 2018, the Turkish economy started experiencing a series of exchange rate depreciation shocks
- Unorthodox monetary policy

Crisis

	Data	Model
Δ Mean Inflation	+25.42 p.p.	+19.36 p.p.
Δ Domestic Currency Share	-20.80 p.p.	-26.73 p.p.
Δ Mean Maturity (DC)	-1.03 years	-1.21 years
Δ Mean Maturity (FC)	-8.87 years	-2.03 years

Δ: post-crisis minus pre-crisis.

Conclusion

1. Analyze data on EMEs' debt issuance and spreads
   - EMEs' domestic currency debt is often of shorter maturity than their foreign currency debt
   - Increased borrowing costs are associated with issuing more
     1.1 foreign currency debt
     1.2 short-term debt (both domestic and foreign currency)
2. Solve a sovereign debt model with rich portfolio choice
   - Formalize & quantify each bond's economic trade-offs
   - Issuing foreign currency debt that is short-term is only optimal during **crises**
   - Issuing domestic currency debt that is long-term is only optimal during **booms**

Thank you!

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Appendix

Data Cleaning

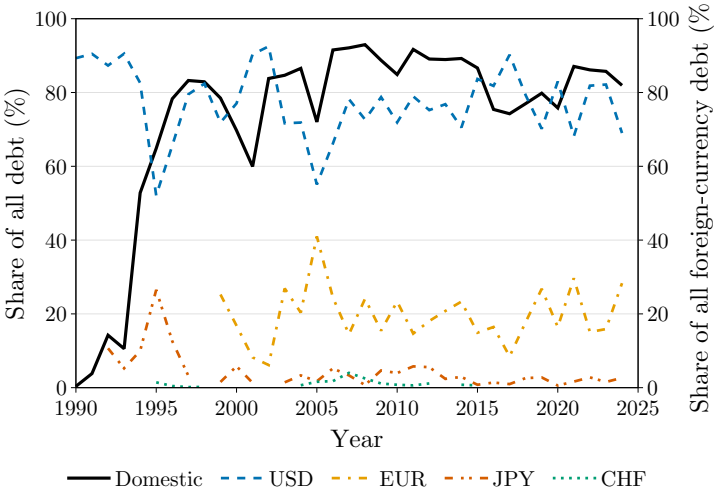
1. Remove bonds that are likely to be bought by domestic investors
   - Bonds with maturities less than 3 months
   - Bonds issued by central banks
2. Remove bonds issued by municipal or sub-national entities

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Summary Statistics of Issuance Data

Country	Sample Period	Number of Bonds	FC Share	Weighted Average Maturity
Angola	Nov. 2005 - Dec. 2024	2,344	9.8%	4.1 years
Argentina	Mar. 1991 - Dec. 2024	874	62.1%	10.9 years
Colombia	Dec. 1990 - Dec. 2024	763	11.8%	11.7 years
Egypt	Dec. 1993 - Dec. 2024	2,715	9.5%	2.0 years
Hungary	Jul. 1993 - Dec. 2024	2,246	13.8%	5.5 years
Indonesia	Jul. 1990 - Dec. 2024	1,146	14.5%	10.6 years
Jamaica	Sep. 1991 - Dec. 2024	778	55.5%	7.3 years
Jordan	Aug. 1993 - Nov. 2024	1,229	17.8%	4.3 years
Mexico	Mar. 1990 - Dec. 2024	2,450	7.2%	4.9 years
Philippines	May 1990 - Dec. 2024	2,066	12.5%	6.5 years
Poland	Dec. 1993 - Nov. 2024	1,242	13.7%	6.4 years
Qatar	Feb. 1999 - Sep. 2024	238	55.0%	10.6 years
Romania	Jun. 1997 - Dec. 2024	888	38.0%	6.9 years
Russia	May 1993 - May 2024	571	12.4%	8.1 years
Tunisia	Sep. 1990 - Dec. 2024	281	21.7%	6.5 years
Türkiye	Mar. 1990 - Dec. 2024	992	15.8%	4.1 years
Ukraine	Mar. 1995 - Dec. 2022	1,813	42.8%	6.1 years
Uruguay	Jan. 1990 - Dec. 2024	348	52.2%	14.9 years
Venezuela	Dec. 1990 - Nov. 2018	1,926	35.0%	7.7 years

Currency Composition of Sovereign Debt



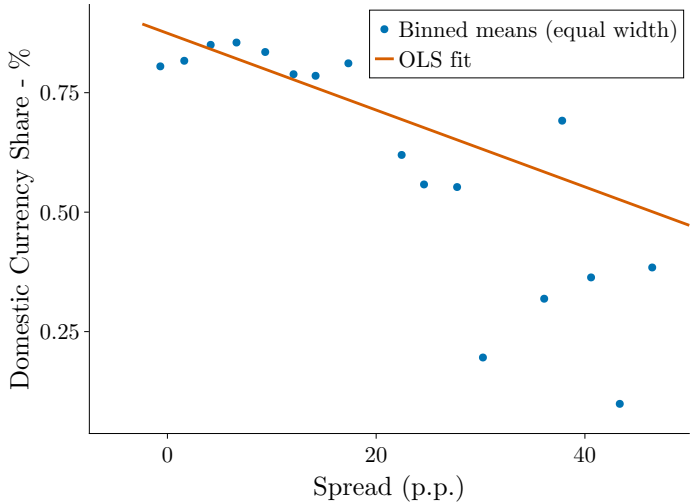
Country	Sample Period	Number of Observations	Source
Colombia	Aug. 2003 - Dec. 2024	5,381	Du and Schreger (2016)
Egypt	Jan. 1997 - Oct. 2024	112	Datastream
Hungary	Jan. 2000 - Dec. 2024	6,450	Du and Schreger (2016)
Indonesia	Feb. 2003 - Dec. 2024	5,488	Du and Schreger (2016)
Jamaica	Jan. 1989 - Oct. 2024	144	Datastream
Mexico	Sep. 2002 - Dec. 2024	5,724	Du and Schreger (2016)
Philippines	Dec. 2000 - Dec. 2024	5,901	Du and Schreger (2016)
Poland	Jan. 2000 - Dec. 2024	6,452	Du and Schreger (2016)
Romania	Jan. 1994 - Oct. 2024	116	Datastream
Russia	Apr. 2006 - Feb. 2022	4,136	Du and Schreger (2016)
Türkiye	Feb. 2005 - Dec. 2024	5,107	Du and Schreger (2016)
Uruguay	Oct. 1991 - Oct. 2024	116	Datastream

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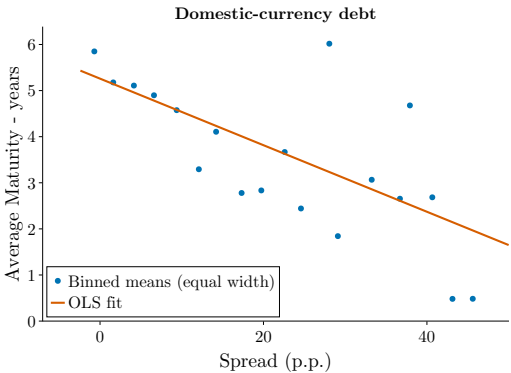
Relationship BW Spreads & Portfolio - Correlations

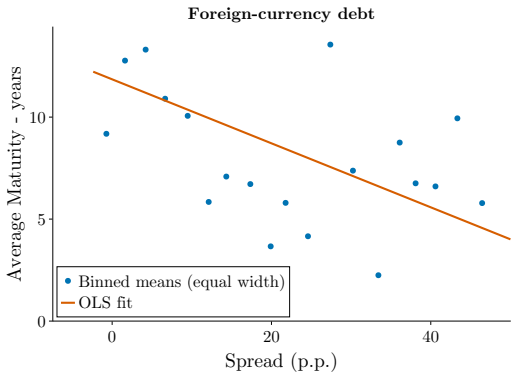
Country	Domestic Share	Maturity (All)	Maturity (Domestic)	Maturity (Foreign)
Colombia	-0.090	-0.055	0.016	-0.388
Egypt	-0.386	0.241	0.136	-0.164
Hungary	-0.091	-0.064	-0.085	0.132
Indonesia	-0.135	-0.051	-0.125	-0.117
Jamaica	-0.136	0.047	-0.023	0.065
Mexico	-0.091	-0.124	-0.097	-0.195
Philippines	-0.147	-0.185	-0.179	-0.436
Poland	0.169	-0.350	-0.318	-0.123
Romania	-0.042	-0.323	-0.392	-0.314
Russia	-0.129	-0.274	-0.306	-0.031
Türkiye	-0.224	-0.067	-0.032	-0.277
Uruguay	-0.384	-0.172	-0.055	-0.275

Relationship Between Spreads & Currency Choice



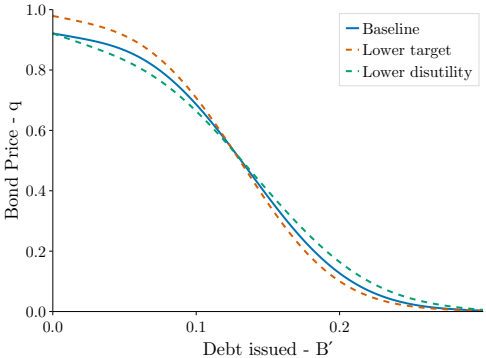
Relationship Between Spreads & Maturity Choice

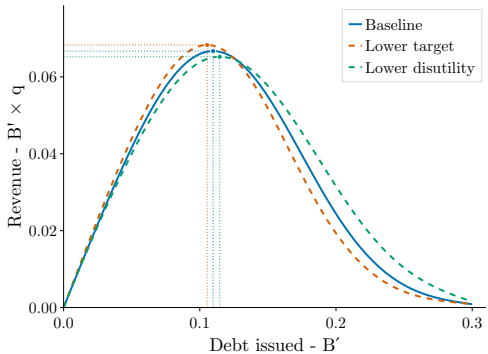




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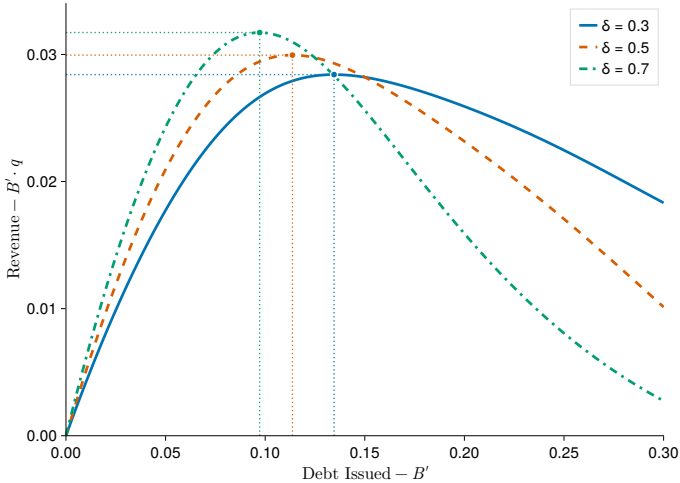
Comparative statics - Currency



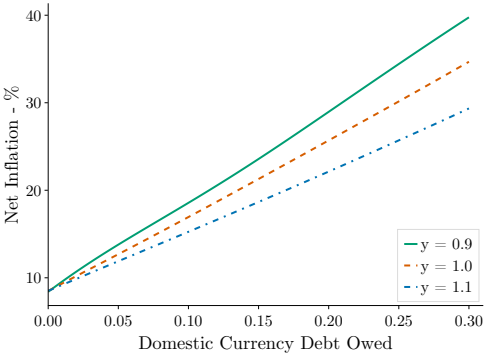


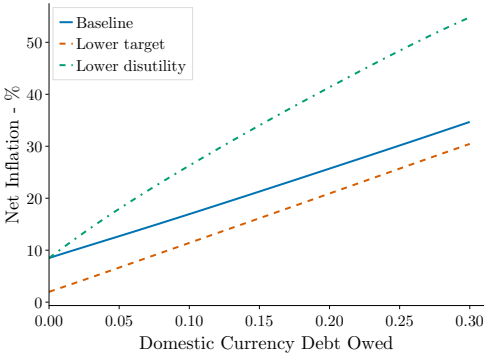
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Comparative statics - Maturity



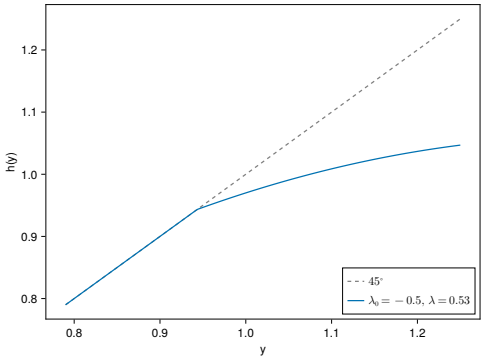
Comparative Statics - Inflation





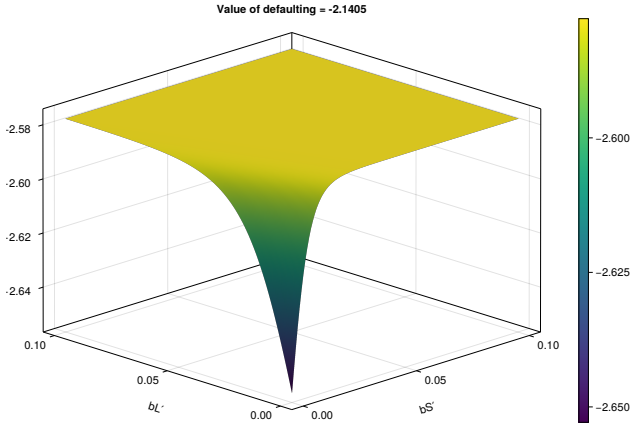
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Output Cost of Defaulting

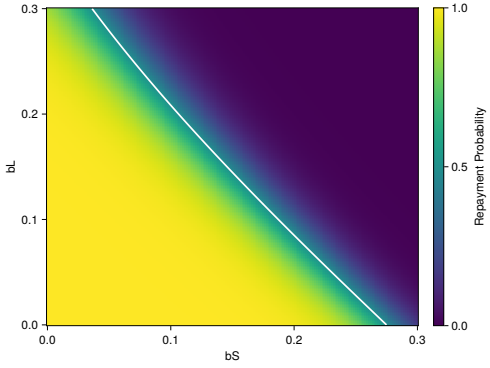


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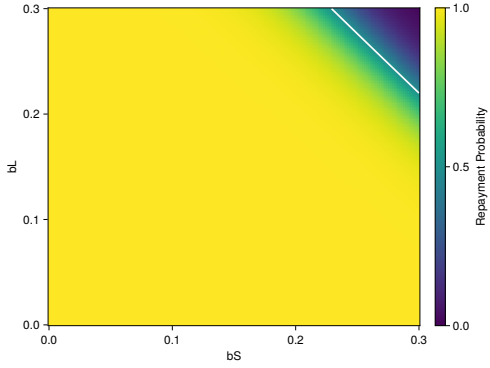
Spurious Optimal Portfolio



Repayment Sets



$y = 1.0$



$y = 1.2$

Taste Shock Details

- Sovereign receives taste shocks to the repayment-default decision

$$\varepsilon^D, \varepsilon^R \sim \text{Gumbel}(-\alpha(\gamma + \ln(2)), \alpha), \quad \gamma \approx 0.57$$

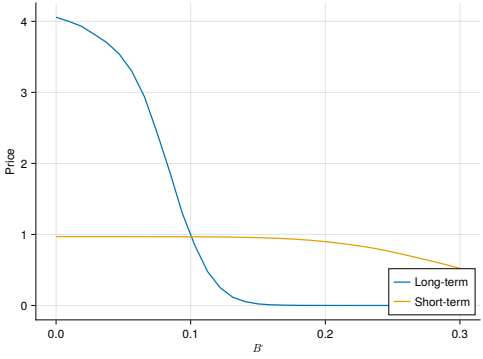
- α: scale parameter
- Ex-ante probability of repayment:

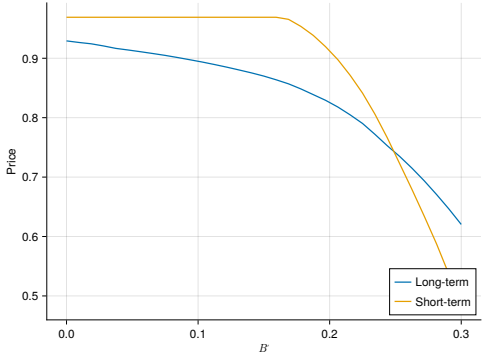
$$\begin{aligned} P(r = 1|\mathbb{B}_t, \mathbb{X}_t) &= P(V^R(\mathbb{B}_t, \mathbb{X}_t) + \varepsilon^R > V^D(\mathbb{X}_t) + \varepsilon^D) \\ &= \frac{\exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right)}{\exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right) + \exp\left(\frac{V^D(\mathbb{X}_t)}{\alpha}\right)} \end{aligned}$$

$$V^O(\mathbb{B}_t, \mathbb{X}_t) = \alpha \ln\left[\exp\left(\frac{V^R(\mathbb{B}_t, \mathbb{X}_t)}{\alpha}\right) + \exp\left(\frac{V^D(\mathbb{X}_t)}{\alpha}\right)\right] - \alpha \ln(2)$$

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Bond prices with and without coupon payments





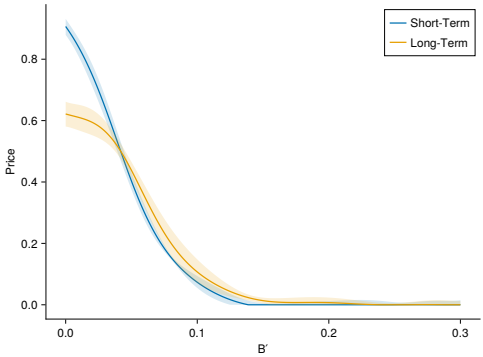
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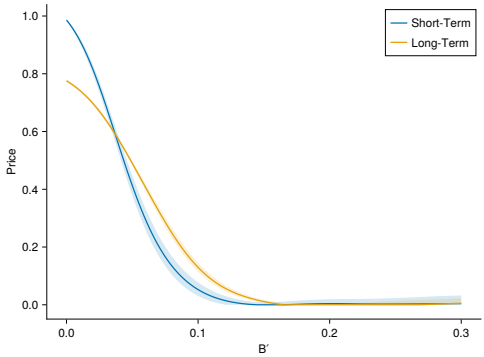
Gaussian Process Regression Details

- Matérn 3/2 kernel
- Noise $= 10^{-6}$
- Automatic Relevance Determination
- Training points
  - $\mu = 5$ Smolyak grid for V^D (145 points)
  - $\mu = 3$ Smolyak grid (389 points) + 311 points via BAL for prices, V^R and $\hat{\Pi}$
  - Repayment frontier for portfolio policy functions

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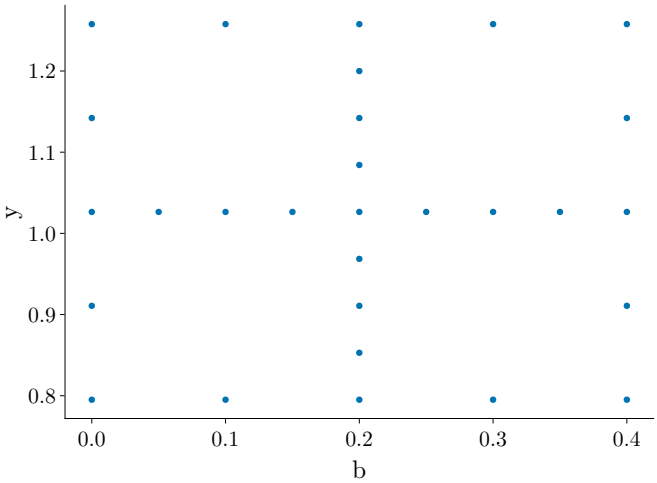
Bayesian Active Learning





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Training Points



Iterative strategy

- Initial guess: limit of finite-horizon game where sovereign buys back all unmatured LT debt
- Iterate backwards until convergence

$$T \to T-1 \to T-2 \dots$$

1. Update pricing functions based on next period's objects
2. Update all value functions and policy function for inflation
3. Use this period's value functions to compute the repayment frontier
4. Generate training points within the frontier and update the policy functions
5. Compute differences and repeat until stationarity

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Türkiye's Inflation

